Habib University
shaping futures

Digital Logic and Design (Lab)

EE/CS 172L/130L T2
Fall Semester 2025

Course Information

Start Date	End Date	Class Location	Meeting Time
Aug 18 2025	Dec 05 2025	W-111	Fri (10:00 AM-01:00 PM)

Course Details	
Credit Hours	01
Course Prerequisites:	None
Hardware Required:	Personal Computer (2-3 GB RAM, 24 GB of Free HDD, Windows 7 or newer)
Software Required:	MS Teams, MS Office, Internet Browser, PDF Reader, Xilinx Vivado WebPack*
Content Area:	This course fulfills the EE, CE & CS Major requirements

Campus Safety	Please read the campus safety policy and COVID-19 protocols if the classes are in-person

Instructor Information

Instructor: Areeba Rajput
Office Location: W-240
Email: areeba.rajput@sse.habib.edu.pk
Office Hours: Tues: 1-2PM

Instructor: Maham Tabassum
Office Location: LG-C001 Circuits & Electronics Lab
Email: maham.tabassum@sse.habib.edu.pk
Office Hours: MW 10:30AM-11:30AM

Course Description

The course is offered as a compulsory course for Computer Science, Computer Engineering and Electrical Engineering programs. The course is intended towards sophomores who want an introduction to the digital logic design. Course starts from first principles of digital information representation and presents the common components and design methodologies needed to design more advanced systems on integrated circuits and programmable logic devices.

Course Aims

The course aims to inculcate following learning goals:

1. Ability to test and implement any Boolean function using discrete logic gate ICs.
2. Ability to analyze digital logic circuits using logic simulation platforms.
3. Ability to write high level description of digital logic circuits using Verilog HDL.
4. Ability to implement any digital logic circuit on FPGAs
5. Ability to perform the functionality of test of Verilog HDL modules.

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

1. CLO: Construct digital logic circuits using Dual-inline Package (DIP) ICs and obtain the correct input-to-output logic

2. CLO: Implement digital logic circuits in Verilog HDL and simulate the Verilog HDL modules using HDL simulation tools

3. CLO: Program FPGA to interface input and output peripherals

4. CLO: Demonstrate ethical values by complying with instructions and practicing organizational skills

Mode of Instruction

The course consists of a 3-hour in-person session of lab per week. The lab content covers applied aspects of the theoretical content covered in lectures and also basic modeling of logical components using Verilog on FPGA.

Engagement & Participation Rules

1. It is expected from students to timely join lab sessions within starting 5 minutes.
2. Students are expected to follow lab safety instructions.
3. Students are expected to be respectful to their class fellows and should adhere to ethical norms. Action will be taken against the use of slangs or any inappropriate language during the sessions which may lead to dropping student from class or dropping from course depending on nature of violence to ethical values.

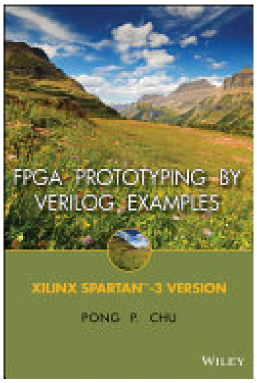
Required Texts and Materials

For each lab, lab manual (.pdf format) will be shared with students through LMS 1 week before lab.

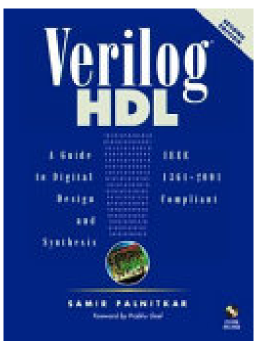
Lab Manual will be editable to add the required simulation results, collected data, reflection on findings and its analysis as instructed in lab manual.

Additional reading material /videos will also be presented on LMS for each lab.

Optional Materials



FPGA Prototyping by Verilog Examples
ISBN: 9781118210611
Authors: Pong P. Chu
Publisher: John Wiley & Sons
Publication Date: 2011-09-20



Verilog HDL
ISBN: 9780130449115
Authors: Samir Palnitkar
Publisher: Prentice Hall Professional
Publication Date: 2003-01-01

Assessments

- Lab work/tasks * 70%
- Lab Project 30%

** Lab tasks include prelab (if any), in-lab, post-lab (if any) tasks and submission of lab report and viva for each lab.*

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)

Letter Grade	GPA Points	Percentage
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60)

Note: [a, b) is a range of numbers from a to b where a is included in the range and b is not.

Attendance Policy

The University's standard Attendance Threshold is a minimum 85% of all sessions (i.e., students must be present in at least 85% of all classes/sessions of a given course) scheduled during the course of the semester. However, instructors may nominate a stricter attendance threshold (but cannot relax it more than the aforementioned 85%). Students must abide by these attendance thresholds set by the relevant faculty. If/Once a student's attendance falls within the aforementioned Attendance Threshold in a given course, they cannot receive credit for that course, and will be removed from the course by the OASR. Students must inform their instructor by the following week's session (of the same day), i.e., within eight (08) days of the discrepant record on PSCS so that the same may be corrected manually on the relevant attendance roster by the instructor. In cases where students are absent from a session on account of medical issues or family emergencies, they must inform/intimate the OASR within ten (10) days.

For more information, please refer to Attendance Policy developed by OASR at Habib University.

Week-Wise Schedule (Tentative)

Week	Lab Title	Remarks
Week - 1 August 18 - 22, 2025	Lab 01: Introduction to Electric Circuits and Lab Equipment	First day of Classes: August 18, 2025 Last day to Drop Course(s): August 23, 2025
Week - 2 August 25 - 29, 2025	Lab 02: Basic Logic Gates	Sessions Last day to Add Course(s): August 25, 2025
Week - 3	Lab 03: Logic Gates and	12th Rabi-ul-Awwal*†: September

September 1 - 5, 2025	Adders	4, 2025
Week - 4 September 8 - 12, 2025	Lab 04: Cascaded Adders	
Week - 5 September 15 - 19, 2025	Lab 05: Introduction to Verilog HDL	
Week - 6 September 22 - 26, 2025	Lab 06: Seven Segment Decoder	
Week - 7 September 29 - October 3, 2025	Lab 07: Integrating Verilog Modules to Design Adder/Subtractor	
Week - 8 October 6 - 10, 2025	Lab 08: Programming BASYS-03 Board	
Week - 9 October 13 - 17, 2025	Lab 09: Multiplexer and Demultiplexer	Mid-Semester Recess†: October 11 & 13 - 14, 2025
Week - 10 October 20 - 24, 2025	Lab 10: Counters and Clock Dividers	
Week - 11 October 27 - 31, 2025	Lab 11: VGA Controller	
Week - 12 November 3 - 7, 2025	Project Week-1	Last Day to Withdraw from Course(s): November 7, 2025
Week – 13 November 10 - 14, 2025	Lab 12: Flip Flop and Finite State Machine	
Week - 14 November 17 - 21, 2025	Project Week-2	
Week – 15 November 24 - 28, 2025	Project Week-3	Conference Days: November 28 - 29, 2025 (No Classes)
Week - 16 December 1 - 5, 2025	Project Final Presentation and Demonstration	Last Day of Classes: December 5, 2025

Final Exam Policy

There will be no final exam.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated,// bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

If you wish to use generative-AI tools to complete any of your assessments, you must first obtain permission from your course instructor. AI generated work will not be accepted in all classes or even all assessments. The instructor's permission is required. If the permission is granted, you should declare its use and properly cite the source of the generated content. Failing to identify AI written or assisted work is academic dishonesty and will be treated as any case of plagiarism by the university.

The principle for academic integrity is that your submissions must be substantially your own work and that any work that is not originally your thought must be identified and credited. If the use of AI tools is prohibited in the course, respect the rules and do not use these tools for assessments. The fundamental purpose of assessment is to learn, synthesize information and explain new connections and interpretations that arise from your secondary research. Be aware that unauthorized use of AI tools for assessments can result in a conduct case being filed. This can have serious consequences for your academic standing and future career opportunities.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Late Submission Policy

Students are expected to submit all the lab tasks on time within submission deadline. If any student submits lab tasks and lab manual late, s/he will be penalized as deduction in 5% per day of maximum allowable marks up to seven days late submissions. No submission will be accepted after seven days of the original deadline unless in exceptional medical cases approved by the Registrar Office.

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 1: Engineering Knowledge: An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

- PLO 3: Design/Development of Solutions: An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

- PLO 5: Modern Tool Usage: An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.

- PLO 8: Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.

CLOs mapped to Program Learning Outcomes (PLOs)				
	Distribution of CLO weightages for each PLO			
	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1	100%	-	-	-
PLO 3	-	100%	-	-
PLO 5	-	-	100%	-
PLO 8	-	-	-	100%

Course Learning Outcomes	Learning domain - level
CLO1: Construct digital logic circuits using Dual-inline Package (DIP) ICs and obtain the correct input-to-output logic	**Psy - 4**
CLO 2: To implementing digital logic circuits in Verilog HDL and simulate the Verilog HDL modules using HDL simulation tools	**Psy - 3**
CLO 3: Program FPGA to interface input and output peripherals	**Psy - 4**
CLO 4: Demonstrate ethical values by complying with instructions and practicing organizational skills	**Aff - 3**

Recording Policy

Only asynchronous and synchronous online sessions will be conducted and recorded via MS Teams. Link to the recordings will be available to all students on Canvas Learning Management System.

Course Assessment Mapping over CLO				
Assessment Item	CLO 1	CLO 2	CLO 3	CLO 4
Lab 01 performance	29.5%			
Lab 01 Viva and report				4.2%
Lab 02 performance	21.3%			
Lab 02 Viva and report				14.8%
Lab 03 performance	24.6%			
Lab 03 Viva and report				10.5%
Lab 04 performance	24.6%			
Lab 04 Viva and report				10.5%
Lab 05 performance		20.2%		
Lab 05 Viva and report				6.3%
Lab 06 performance		19.0%		
Lab 06 Viva and report				8.4%

Lab 07 performance		**21.4%**		
Lab 07 Viva and report				**4.2%**
Lab 08 performance			**12.0%**	
Lab 08 Viva and report				**4.2%**
Lab 09 performance		**17.9%**		
Lab 09 Viva and report				**10.5%**
Lab 10 performance		**21.4%**		
Lab 10 Viva and report				**4.2%**
Lab 11 performance			**11.3%**	
Lab 11 Viva and report				**6.3%**
Lab 12 performance		**10.6%**		
Lab 12 Viva and report				**8.4%**
Project			**66.1%**	**7.2%**

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other’s opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar’s Office.